

🛑 ASSUMPTIONS IN MATH (1.2 Methods) 🛑

What is an Assumption?

- A condition we **accept as true** so we can solve a messy real-life problem.
- *Real life is unpredictable; math needs rules!*

Why do we need them?

- To make a mathematical calculation possible.
- If we don't assume *something*, we can't calculate *anything*.



EXAMPLES IN ACTION

Scenario A: Biking a Trail

- **The Math:** A mountain biker rides a 15 km track at an average speed of 10 km/hr. It will take exactly 1.5 hours.
- **The Assumption:** The rider maintains a *constant speed* the entire time.
- **The Reality Check:** We know they will likely slow down on uphill climbs, speed up on descents, or get tired.

Scenario B: Buying Gear for the Team

- **The Math:** Buying 5 new basketballs for the KVC squad costs \$100 (so \$20 each). Therefore, buying 20 basketballs will cost \$400.
- **The Assumption:** The price scales proportionally and there are *no bulk discounts* or shipping limits.



HOW TO WRITE IT IN THE EXAM (Get the grade!)

To get Merit/Excellence, you can't just state the assumption. You **MUST** explain **how it affects your answer**.

Use this formula:

"I am assuming that [insert assumption here]. If this is not true, and [give an alternative], then my final answer would be [higher / lower / invalid]."

Example Answer:

*"I calculated the bike ride will take 1.5 hours. I am assuming the rider travels at a constant speed of 10 km/hr without stopping. If they get tired or stop for water, their average speed will drop, and the total time taken will be **higher** than 1.5 hours."*



QUICK CHECKLIST FOR STUDENTS

Before finishing a problem, ask yourself:

1. Did I assume a **constant rate/speed**?
2. Did I assume **no outside factors** (like weather, traffic, fatigue)?
3. Did I assume **proportional costs** (no sales, no bulk deals)?
4. Did I link the assumption back to the **final number**?